

PADHAISHURU

GATE Preparation Platform

GATE GG

Geology & Geophysics

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	6 subjects

Difficulty Distribution

Easy: 27 questions (42%)
Moderate: 30 questions (46%)
Difficult: 8 questions (12%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE GG — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

1M [-0.33]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q2.

1M [-0.33]

[Engineering Mathematics]

Mean of 1,2,3,4,5:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q3.

1M [-0.33]

[Geology]

Study of fossils:

- (A) Paleontology
- (B) Petrology
- (C) Mineralogy
- (D) None

Q4.

1M [-0.33]

[Geology]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q5.

1M [-0.33]

[Geology]

Law of superposition by:

- (A) Steno
- (B) Lyell
- (C) Darwin
- (D) Hutton

Q6.

1M [no -ve]

[Geophysics]

P-waves are:

Enter your answer here

Q7.

1M [-0.33]

[Geophysics]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q8.

1M [-0.33]

[Geophysics]

Earth's magnetic field generated by:

- (A) Core
- (B) Crust
- (C) Mantle
- (D) Atmosphere

Q9.

1M [no -ve]

[Remote Sensing & GIS]

Landsat uses:

*Enter your answer here***Q10.**

1M [-0.33]

[Petrology]

Igneous rock from:

- (A) Sediment
- (B) Cooling magma
- (C) Metamorphism
- (D) Organic

Q11.

1M [-0.33]

[Geophysics]

P-waves are:

- (A) Primary
- (B) Secondary
- (C) Surface
- (D) Love

Q12.

1M [-0.33]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q13.

1M [-0.33]

[Petrology]

Igneous rock from:

- (A) Sediment
- (B) Cooling magma
- (C) Metamorphism
- (D) Organic

Q14.

1M [no -ve]

[Petrology]

Igneous rock from:

*Enter your answer here***Q15.**

1M [no -ve]

[Engineering Mathematics]

Mean of 1,2,3,4,5:

*Enter your answer here***Q16.**

1M [-0.33]

[Geology]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q17.

1M [no -ve]

[Engineering Mathematics]

Mean of 1,2,3,4,5:

*Enter your answer here***Q18.**

1M [no -ve]

[Petrology]

Igneous rock from:

*Enter your answer here***Q19.**

1M [-0.33]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q20.

1M [no -ve]

[Geology]

Hardest mineral:

*Enter your answer here***Q21.**

1M [-0.33]

[Geophysics]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q22.

1M [-0.33]

[Geology]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q23.

1M [-0.33]

[Petrology]

Igneous rock from:

- (A) Sediment
- (B) Cooling magma
- (C) Metamorphism
- (D) Organic

Q24.

1M [-0.33]

[Geology]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q25.

1M [-0.33]

[Geophysics]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q26.

1M [no -ve]

[Geophysics]

Gravity anomaly due to:

*Enter your answer here***Q27.**

1M [no -ve]

[Geology]

Hardest mineral:

*Enter your answer here***Q28.**

1M [no -ve]

[Geology]

Hardest mineral:

*Enter your answer here***Q29.**

1M [-0.33]

[Geology]

Law of superposition by:

- (A) Steno
- (B) Lyell
- (C) Darwin
- (D) Hutton

Q30.

1M [-0.33]

[Engineering Mathematics]

Mean of 1,2,3,4,5:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q31.

2M [-0.66]

[Geophysics]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q32.

2M [-0.66]

[Geophysics]

P-waves are:

- (A) Primary
- (B) Secondary
- (C) Surface
- (D) Love

Q33.

2M [no -ve]

[Geology]

Law of superposition by:

*Enter your answer here***Q34.**

2M [-0.66]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q35.

2M [-0.66]

[Geology]

Study of fossils:

- (A) Paleontology
- (B) Petrology
- (C) Mineralogy
- (D) None

Q36.

2M [no -ve]

[Remote Sensing & GIS]

Landsat uses:

*Enter your answer here***Q37.**

2M [-0.66]

[Remote Sensing & GIS]

Landsat uses:

- (A) Optical
- (B) Radar
- (C) Both
- (D) None

Q38.

2M [no -ve]

[Petrology]

Igneous rock from:

*Enter your answer here***Q39.**

2M [no -ve]

[Geology]

Hardest mineral:

Enter your answer here

Q40.

2M [no -ve]

[Geophysics]

P-waves are:

*Enter your answer here***Q41.**

2M [no -ve]

[Remote Sensing & GIS]

Landsat uses:

*Enter your answer here***Q42.**

2M [-0.66]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q43.

2M [no -ve]

[Petrology]

Igneous rock from:

*Enter your answer here***Q44.**

2M [-0.66]

[Geology]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q45.

2M [-0.66]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q46.

2M [-0.66]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q47.

2M [-0.66]

[General Aptitude]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Q48.

2M [-0.66]

[Geology]

Law of superposition by:

- (A) Steno
- (B) Lyell
- (C) Darwin
- (D) Hutton

Q49.

2M [-0.66]

[Geology]

Law of superposition by:

- (A) Steno
- (B) Lyell
- (C) Darwin
- (D) Hutton

Q50.

2M [-0.66]

[Petrology]

Igneous rock from:

- (A) Sediment
- (B) Cooling magma
- (C) Metamorphism
- (D) Organic

Q51.

2M [no -ve]

[General Aptitude]

Choose similar to 'volcanic':

Enter your answer here

Q52.

2M [-0.66]

[Remote Sensing & GIS]

Landsat uses:

- (A) Optical
- (B) Radar
- (C) Both
- (D) None

Q53.

2M [-0.66]

[Geophysics]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q54.

2M [-0.66]

[Geology]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q55.

2M [-0.66]

[Geophysics]

P-waves are:

- (A) Primary
- (B) Secondary
- (C) Surface
- (D) Love

Q56.

2M [-0.66]

[Petrology]

Igneous rock from:

- (A) Sediment
- (B) Cooling magma
- (C) Metamorphism
- (D) Organic

Q57.

2M [no -ve]

[Remote Sensing & GIS]

Landsat uses:

*Enter your answer here***Q58.**

2M [no -ve]

[Geology]

Hardest mineral:

*Enter your answer here***Q59.**

2M [no -ve]

[Petrology]

Igneous rock from:

Enter your answer here

Q60.

2M [-0.66]

[Petrology]

Igneous rock from:

- (A) Sediment
- (B) Cooling magma
- (C) Metamorphism
- (D) Organic

Q61.

2M [no -ve]

[Geophysics]

Earth's magnetic field generated by:

*Enter your answer here***Q62.**

2M [no -ve]

[Remote Sensing & GIS]

Landsat uses:

*Enter your answer here***Q63.**

2M [no -ve]

[Geology]

Hardest mineral:

*Enter your answer here***Q64.**

2M [no -ve]

[Geophysics]

Earth's magnetic field generated by:

*Enter your answer here***Q65.**

2M [no -ve]

[Geology]

Hardest mineral:

Enter your answer here